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25 May 1990

SUPERSEDING
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MILITARY SPECIFICATION

COMPUTER, RANGE CP-888/VPS-2: 10548339

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Computer, Range CP-888/VPS-2, hereinafter called the computer. The computer is part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel, Manufacture and - Inspection, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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MIL-P-50317 - Protector, Over Voltage: 10548031 (+5 Volt Filter)
MIL-R-50318 - Regulator, Voltage: 10548034
MIL-R-50319 - Regulator, Voltage: 10548043
MIL-G-50322 - Generator, Digital: 10548613
MIL-C-50344 - Counter, Electronic: 10548009
MIL-R-50651 - Radar Set AN/VPS-2, 10549058 (Range Only Radar)
MIL-C-50652/1 - Modulator Trigger Generator: 9355881
MIL-C-50652/2 - Computer Timing Generator: 10548007
MIL-C-50652/3 - RCFD Decoder: 10548119
MIL-C-50652/4 - Range Counter: 10548046
MIL-C-50652/5 - RCFD Generator: 10548016
MIL-C-50652/6 - Range Register A: 10548122
MIL-C-50652/7 - Range Register B: 10548021
MIL-C-50652/8 - Search Track and Gain Control: 10548148
MIL-C-50652/9 - Range Rate Counter A: 10548125
MIL-C-50652/10 - Range Rate Counter B: 10548012
MIL-C-50652/11 - Rate Scaler: 10548049
MIL-C-50652/12 - Rate Integrator: 10548086
MIL-C-50652/13 - Phase Detector: 10548133
MIL-C-50652/14 - Range D/A Converter: 10548359
MIL-C-50652/15 - Range Rate D/A Converter: 10548330
MIL-C-50652/16 - Control Amplifier: 10548450
MIL-C-50652/17 - Fine/Fine Delay: 10548638
MIL-C-50652/18 - Universal Board: 10549234

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548339 - Computer, Range CP-888/VPS-2

PACKAGING DATA SHEET

PS 10548339 - Packaging of Computer, Range CP-888/VPS-2

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548339, all associated drawings, and with all requirements specified in applicable specifications and standards. The subassemblies of the computer shall have been inspected and accepted in accordance with their respective drawings or specifications prior to their installation in the computer. The following subassemblies are covered by the indicated specifications:

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UNIT NO.	NAME	SPECIFICATION
A1	Generator Digital: 10548613	MIL-G-50322
A2	Counter Electronic: 10548009	MIL-C-50344
A3	Modulator Trigger Generator: 9355881	MIL-C-50652/1
A4	Computer Timing Generator: 10548007	MIL-C-50652/2
A5	RCFD Decoder: 10548119	MIL-C-50652/3
A6	Range Counter: 10548046	MIL-C-50652/4
A7	RCFD Generator: 10548016	MIL-C-50652/5
A8	Range Register A: 10548122	MIL-C-50652/6
A9	Range Register B: 10548021	MIL-C-50652/7
A10	Search, Track and Gain Control: 10548148	MIL-C-50652/8
A11	Range Rate Counter A: 10548125	MIL-C-50652/9
A12	Range Rate Counter B: 10548012	MIL-C-50652/10
A13	Rate Scaler: 10548049	MIL-C-50652/11
A14	Rate Integrator: 10548086	MIL-C-50652/12
A15	Phase Detector: 10548133	MIL-C-50652/13
A16	Range D/A Converter: 10548359	MIL-C-50652/14
A17	Range Rate D/A Converter: 10548330	MIL-C-50652/15
A18	Protector, Overvoltage: 10548031 (+5 volt filter)	MIL-F-50317
A19	Regulator, Voltage: 10548034	MIL-R-50318
A20	Regulator, Voltage: 10548043	MIL-R-50319
A21	Control Amplifier: 10548450	MIL-C-50652/16
A22	Fine/Fine Delay: 10548638	MIL-C-50652/17
A23	Universal Board: 10549234	MIL-C-50652/18

Where applicable, the computer shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

3.3 Environmental service conditions.

3.3.1 Vibration. The computer shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ±10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ±10%.

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3.3.2 Shock. The computer shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 ± 10% g's, 11 ± 10% milliseconds, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The computer shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The computer shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The computer shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 Degrees F and +155 Degrees F.

3.3.4.2 Non-operating (storage). The computer shall not suffer permanent damage when subjected to the constant temperature extremes of -60 Degrees F for a period of 24 hours and +160 Degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The computer shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the range computer.

3.5 Performance. For the requirements of 3.5.1 through 3.5.26 the following conditions shall apply:

a. Logic 0: 0.0 ±0.5 volts dc (Vdc) level.
b. Logic 1: +3.5 ±1.0, Vdc level.
c. Positive pulse: a change in dc level from 0.0 ±0.5 V to +3.5 ±1.0 V and return to 0 V.
d. Negative pulse: a change in dc level from +3.5 ±1.0 V to 0.0 ±0.5 V and return to +3.5 V.
e. Switch 4S2 (Mode): in test position.
f. Switch 4S3 (Accel.): in negative position.
g. Switch 4S4 (Return): in no target position.
h. Switch 4S5 (Range): in run position.
i. Switch 4S6 (Rate): in normal position.
j. Delays in signals when specified shall be from the leading edges of each signal.
k. All pulse widths are specified at the 50 percent points.

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3.5.1 Voltage inputs and distribution. With the input voltages applied as specified in Table I, the internal voltages for distribution to the assemblies in the computers shall be as specified in Table II.

TABLE I. Input voltages

INPUT VOLTAGE (Vdc)	CHARACTERISTICS	TERMINALS (Vdc)
-33	-33.00 ±0.25	J1-a(-), J1-b(ret)
+33	+33.00 ±0.25	J1-Y(+), J1-Z(ret)
-23	-23.00 ±0.25	J1-j(-), J1-c(ret)
+23	+23.00 ±0.25	J1-h(+), J1-c(ret)
+6.2	+6.20 ±0.50	J1-e(+), J1-g(ret) J1-d(+), J1-f(ret)
+24	+24.0 ±0.5	J1-T(+), J1-U(Ret)

TABLE II. Distributed voltages

VOLTAGE (Vdc)	CHARACTERISTICS (Vdc)	TERMINAL TEST POINT
-25	-25.0 ±0.5	E1(-), E2(ret)
+25	+25.0 ±0.5	E3(+), E4(ret)
-23	-15.0 ±0.5	E5(-), E6(ret)
+23	+15.0 ±0.5	E8(+), E7(ret)
+5	+5.0 ±0.5	E9(+), E10(Ret) W1(+), W2(ret)

3.5.2 Mid range lamp, 4DS1. With voltages applied as specified in Table I and no input signals applied, the mid-range lamp 4DS1 shall be dimly lit.

3.5.3 Mid range switch, 4S1. With +24 Vdc applied to 4J1-T and with "PRESS TO TEST" switch 4S1 depressed, there shall be +24 Vdc present at 4J1-R and 4J3-C.

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3.5.4 7.5 MHz clock output. With a 60.0 ±5.0 megahertz (MHz) signal having an amplitude between the limits of 2.5 to 4 V peak-to-peak applied to J4-A, a 7.5 ±0.6 MHz clock signal shall be present at TP38 on the computer. The clock signal shall have a period of 133 ±10 nanoseconds (ns) with an amplitude of 4.0 V peak-to-peak or greater.

3.5.5 PRF RESET signal. The PRF RESET signal at TP20 shall be a negative pulse, 67 ±25 ns wide, occurring every 50 ±5 microseconds (us).

3.5.6 PRF signal. The PRF signal at TP19 shall be a negative pulse having the same characteristics as specified in 3.5.5.

3.5.7 PRF CAL signal. With 4S1, mid range C-1 switch depressed, the PRF CAL signal at TP19 shall be two negative pulses. Each pulse shall be 67±25 ns from each other. The first pulse shall be delayed 16 ±2 us from the signal specified in 3.5.5.

3.5.8 PRF signal. The PRF signal at TP31 shall be a negative pulse, 0.6 ±0.1u s wide, occurring every 50 ±5u s. With switch 4S1 depressed the negative pulse at TP31 shall be 0.8 ±0.1μ s wide, occurring every 50 ±5μ s, and be delayed 16 ±2μ s from the signal specified in 3.5.5.

3.5.9 $\overline{T1}$ signal. The $\overline{T1}$ signal at TP21 shall be a positive pulse, 0.75 ±0.10u s wide occurring every 50 ±5μ s, and be delayed 34 ±4μ s from the signal specified in 3.5.5.

3.5.10 MOD T_0 trigger signal. The MOD T_0 trigger signal at TP45 and 4J4-E shall be a positive pulse of not less that 17 volts peak and having a maximum rise time of 100 ns into a 150 ohm load. Any pulse ringing shall not recross a 15.5 volt threshold. The signal TP45 shall be delayed 150 ±30 ns from the signal specified in 3.5.8 (delay shall be measured from rising edge to rising edge at 50% points). The signal shall show no changes when switch 4S1 is depressed.

3.5.11 T2 signal. The T2 signal at TP34 shall be as specified in 3.5.9 except the delay shall be 1.0 ±0.1μ s from the T1 signal specified in 3.5.9.

3.5.12 T3 signal. The T3 signal at TP23 shall be as specified in 3.5.8 except the delay shall be 1.0 ±0.1μs from the T2 signal specified in 3.5.11.

3.5.13 $\overline{T11}$ signal. The T11 signal at TP32 shall be a negative pulse, 0.75 ±0.10 s wide occurring every 50 ± 5μ s, and delayed 43 ±3μ s from the PRF RESET signal specified in 3.5.5.

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3.5.14 $\overline{\text{T1/250 signal}}$. The T1/250 signal at TP9 shall be a complementary pulse (negative pulse) to the T1 signal as specified in 3.5.9 except they shall occur every 4.0 ±0.5 ms.

3.5.15 $\overline{\text{T3/4KC signal}}$. The T3/4KC signal at TP10 shall be a coincident pulse to the T3 signal as specified in 3.5.12 except they shall occur every 250 ±25u s.

3.5.16 $\overline{\text{200M signal}}$. With switch 4S4 in the target position, the 200M signal at TP11 shall be a logic 1 level. With switch 4S4 in the target position and 24 Vdc applied between 4J1-G(+) and 4J1-H(return), the 200M signal at TP11 shall be a logic 0 level.

3.5.17 T3 T10 serial and R serial signals. With the switches 4S2 through 4S6 sequentially placed as specified in Table III, the T3 T10 serial and R serial signals at TP17 and TP18 shall be as specified in Table III. Signal waveform characteristics are shown in Figure 1. When going from step 1 to step 2 of Table III and the signal at TP18, as shown in Figure 1b, changes to that shown in Figure 1c, all eight time slots of the signal shall be counting, i.e., no pulses shall remain always present or always missing.

TABLE III. Placement of switches

STEP	4S2 MODE	4S3 ACCEL	4S4 RETURN	4S5 RANGE	4S6 RATE	SIGNAL CHARACTERISTICS TP17	TP18
1.	Test	Pos.	Tgt.	Run	Normal	Fig. 1a	Fig. 1b
2.	Test	Neg.	Tgt.	Run	Normal	Fig. 1a	Fig. 1c
3.	Test	Pos.	Tgt.	Run	Reset	Fig. 1a	Logic 0
4.	Test	Neg.	Tgt.	Run	Reset	Fig. 1a	Logic 0
5.	Test	Pos.	No tgt.	Run	Normal	Fig. 1a	Logic 0
6.	Test	Neg.	No tgt.	Run	Normal	Fig. 1a	Logic 0

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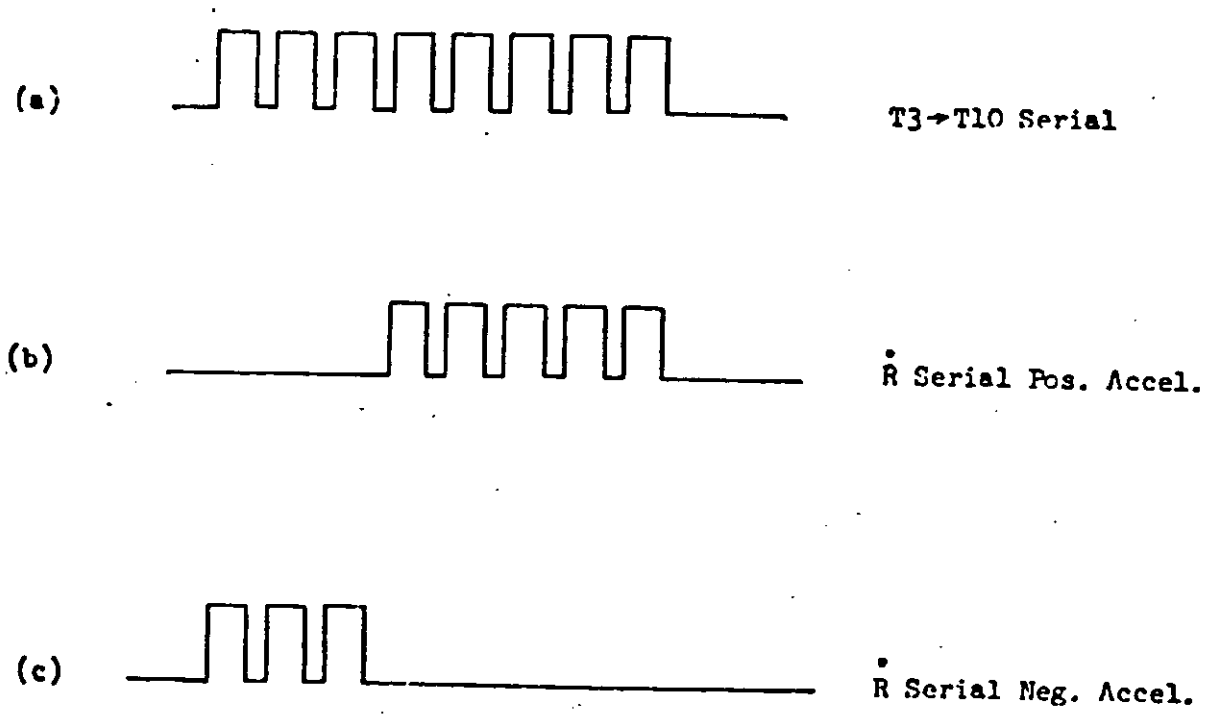


FIGURE 1. TP3 → TP10 and $\dot{R}$ serial signals.

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3.5.18 TRACK signal.

3.5.18.1 No target condition. With switches 4S2 through 4S6 placed in positions as specified in 3.5, the TRACK signal at TP15 and TP24 shall be a logic 0 level.

3.5.18.2 Target condition. When switch 4S4 is changed from the no target position to the target position, the TRACK signal at TP15 and TP24 shall change from a logic 0 level to a logic 1 level.

3.5.18.3 Target fade (short term). When switch 4S4 is changed from the target position to the no target position, the TRACK signal at TP15 and TP24 shall change from a logic 1 level to a logic 0 level after 1.00 ±0.20 second(s).

3.5.18.4 Target fade (long term). When switch 4S6 is placed in the reset position and simultaneously switch 4S4 is changed from the target position to the no target position, the TRACK signal at TP15 and TP24 shall change from a logic 1 level to a logic 0 level after 3.00 ±0.25 s.

3.5.18.5 Target lock-on signal. When switch 4S4 is changed from the no target position to the target position, the target lock-on signal between 4J5-C(signal) and 4J-D(signal return) shall change from a logic 1 level to a logic 0 level.

3.5.19 MID RANGE lamp 4DS1. With switch 4S4 in the NO TARGET position, the mid range lamp 4DS1 shall be flashing brightly at 1-second intervals.

3.5.20 E/L COURSE signal. The E/L COURSE signal at TP33 shall be a single positive-going step function, whose leading edge moves from less than 2.0μ s to 33 ±3μ s, with respect to the PRF signal at TP31, at 1-second intervals.

3.5.21 PRE RANGE trigger. The PRE RANGE trigger signal at TP36 and 4J4-B shall be a single positive-going step function whose leading edge moves in synchronization with, and less than 2u s behind, the E/L COARSE signal at TP33.

3.5.22 E/L TRIGGER.

3.5.22.1 E/L TRIGGER characteristics. The E/L trigger signals shall be present at TP43, TP44, and 4J4-D, and have the characteristics as specified in Table IV.

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TABLE IV. E/L trigger characteristics

TERMINAL	CHARACTERISTICS
TP43	Positive pulse 0.4 to 1.5μ s wide, and move from a delay of less than 2μ s to a delay of 33 ±3μ s with respect to signal at TP31.
TP44	Positive pulse 0.2 to 0.6μ s wide, and move with the same delays as specified for TP43.
4J4-D	Positive pulse with the same characteristics as signal at TP44 except delayed by 1.0 ±0.3μ s from signal at TP44.

3.5.22.2 E/L TRIGGER movements. When the switches 4S3 through 4S6 are sequentially positioned as specified in Table V, the E/L trigger signals at TP43 and TP44 shall have the characteristics as specified in Table IV except the direction and times of movement shall be as specified in Table V.

TABLE V. E/L trigger movement

	SWITCH POSITIONS				SIGNAL CHARACTERISTICS
STEP	4S5	4S3	4S4	4S6	TP43, TP44
1	Run	Neg.	No tgt.	Normal	Left to right motion at 1 s rate.
2.	Freeze	Neg.	No tgt.	Normal	No motion.
3.	Freeze	Pos.	Tgt.	Normal	No motion.
4.	Run	Pos.	Tgt.	Normal	Left to right motion at a 17 ±1 s rate.
5.	Run	Neg.	Tgt.	Normal	Right to left motion at a 17 ±1 s rate.
6.	Run	Toggle switch between pos. & neg. several times	Tgt.		Pulse motion shall be faster (almost twice as fast) for the first couple of seconds after each toggle of 4S4, then resume the 17 s rate.

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TABLE V. E/L trigger movement (Cont.)

SWITCH POSITIONS				SIGNAL CHARACTERISTICS
STEP	4S5	4S3	4S4	4S6	TP43, TP44
7	Run	Same as Step 6.	Tgt.	Reset momentarily for each toggle.	Pulse Motion shall cease until 4S6 is placed back to normal, then pulse motion shall be double for first couple of seconds, then resume the 17 s rate.

3.5.22.3 E/L TRIGGER leading edges. The E/L trigger signals at TP43, TP44, and 4J4-D, shall have leading edges with the characteristics as specified in Table VI; switches 4S2 through 4S6 shall be as specified in 3.5.

TABLE VI. E/L trigger leading edges

TEST POINT OR TERMINAL	TEST POINT FOR SYNCHRONIZATION	CHARACTERISTIC
TP43	TP33	Sixteen leading edges with delay limits (ns) as specified below.

	NLT	NMT
1	7.1	9.7
2	14.0	19.0
3	21.0	29.0
4	28.0	38.0
5	35.0	48.0
6	43.0	58.0
7	50.0	67.0
8	57.0	77.0
9	64.0	87.0
10	71.0	96.0
11	78.0	105.0
12	85.0	115.0
13	92.0	125.0
14	99.0	134.0
15	105.0	143.0

TP44	TP43	Five leading edges evenly spaced 1.7 ±0.2 ns. The total duration from first to last leading edge shall be 6.5 ± 0.5 ns.
4J4-D	TP43	Five leading edges with the same characteristics as for TP44 except with the delay as specified for 4J4-D in Table IV.

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3.5.23 R> +10M/S signal. When the steps of Table VII are sequentially performed, the R> ±10M/S signal at TP42 shall be as specified in Table VII under the following conditions.

a. 4S2 placed in Test.
b. 4S3 placed as specified in Table VII.
c. 4S4 placed in Target.
d. 4S5 placed in Run.
e. 4S6 as specified in Table VII.

TABLE VII. R > +10M/S signal

STEP	4S3	4S6	TP42 SIGNAL CHARACTERISTICS
1	Neg.	Run	Logic 0 level.
2	Neg.	Reset	Logic 1 level
3	Note	Run	Momentary transition from logic 0 to logic 1 and back to logic 0 level again each time 4S3 is toggled.

NOTE: Toggle 4S3 between the negative and positive positions several times.

3.5.24 Range rate analog signals. When the switches 4S2 through 4S6 are sequentially positioned as specified in Table IX, the range rate signals shall be as specified in Table VIII.

TABLE VIII. Range rate analog signals

REQUIREMENT	CHARACTERISTIC	TERMINALS	
		COMPUTER	TEST FIXTURE
+310M/S ref.	Adjustable to +10.000 ±0.050 Vdc	4J5-K	TP-2
-310M/S ref.	-10.000 ±0.050 Vdc	4J5-G	TP-3
Range rate output	As specified in Table IX	4J5-H	TP-1
Range rate return	Return side for rate signals	4J5-F	TP-4

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TABLE IX. Range rate output signal

STEP	4S2	4S3	4S4	4S5	4S6	OUTPUT SIGNAL
1	Test	Pos.	Tgt.	Run	Run	Adjustable to +9.775 ±0.005 Vdc. Note 1.
2	Test	Neg.	Tgt.	Run	Run	Adjustable to -9.775 ±0.005 Vdc. Note 1.
3	Test	Neg	Tgt.	Run	Reset	0.0 ±0.1 Vdc.
4	Test	Neg.	No tgt.	Run	Run	-6.4 ±0.1 Vdc.

NOTE 1: Output can vary an additional ±1% for environmental testing.

3.5.25 Range analog signals. When switches 4S2 through 4S6 are sequentially positioned as specified in Table XI the range analog signals shall be as specified in Table X.

TABLE X. Range analog signals

REQUIREMENT	CHARACTERISTICS	TERMINALS	
		COMPUTER	TEST FIXTURE
5000M range ref.	+20.0 ±0.1 V	4J5-J	TP-5
Range output	As specified in Table XI.	4J5-A	TP-6
Range return	Return side for signals.	4J5-B	TP-7

TABLE XI. Range output signal

STEP	4S2	4S3	4S4	4S5	4S6	OUTPUT SIGNAL
1	Test	Pos.	No tgt.	Run	Run	+6.4 ±0.1 Vdc
2	Test	Pos.	Tgt.	Run	Run	.0 ±0.1 Vdc rising to +20.0 ±0.1 Vdc, taking 17 ±1 s to complete.
3	Test	Neg.	Tgt.	Run	Run	+20.0 ±0.1 Vdc falling to 0.0±0.1 Vdc, taking 17 ±1 s to complete.

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TABLE XI. Range output signal (Cont.)

STEP	4S2	4S3	4S4	4S5	4S6	OUTPUT SIGNAL
4	Test	Neg.	Tgt.	Freeze	Run	DC level stops changing and remains stationary.
5	Test	Neg.	No tgt.	Run	Reset	+6.4 ±0.1 Vdc.
6	Test	Neg.	Tgt.	Run	Reset	Same as specified in Step 4.
7	Test	Neg.	Tgt.	Run	Run	Dc level shall continue to fall.

3.5.26 Doppler signals response. The computer shall respond to the application of sum and difference signals sequentially applied as specified in Table XII and with the following conditions.

a. Sum and difference input signal characteristics as specified in Table XIII.

b. The E/L trigger at TP44 shall move as specified in Table XIII, between the delay limits of less than 2μ s to 33 ±3μ s with respect to the signal at TP45.

c. The ECM relay (4K3) shall provide contact closures between terminals 4J4-B and as specified in Table XIII.

d. The TGT lock-on relay (4K2) shall provide contact closures between terminals 4J4-D and as specified in Table XIII.

e. The doppler audio output test signal shall be as specified in Table XIII for the conditions of Step 4 of Table XII.

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TABLE XII. Doppler signal response

STEP PHASE	TP44 SIGNAL	RELAY CLOSURES ECM	LOCK-ON	SIGNAL INPUTS SUM	DIFFERENCE	
1	Left to right, 1 s	4J1-A	4J1-C	No signal	No signal	
2	Left to right, 1 s	4J1-A	4J1-C	No signal	1.5 V	170°
3	Right to left, 17 s	4J1-A	Note 1	0.2 V	0.75 V	170°
4	Left to right, 17 s	4J1-A	4J1-w	0.2 V	0.75 V	10°
5	Stationary (note 2)	4J1-A	4J1-w	0.2 V	0.75 V	10°
6	Left to right, 17 s	4J1-A	4J1-w	0.2 V	0.75 V	10°
7	Stationary (note 2)	Note 3	4J1-w	1.1 V	0.75 V	10°

NOTES: 1. Adjustable to point where contact closure for terminal 4J1-w shall occur when the 0.2V sum input level is reached.

2. With the RATE switch (4S6) in reset position the E/L trigger signal shall stop moving and remain stationary.

3. Adjustable to point where contact closure for terminal 4J1-V shall occur when the 1.1 V sum input level is reached.

TABLE XIII. Doppler input and output signals

REQUIREMENT	CHARACTERISTIC	TERMINAL
Sum signal	Reference phase: 1.0 ±0.1 kHz sine wave, amplitude in peak-to-peak volts as specified in Table XII and as measured between TP47 (signal) and TP46 (signal return) on the computer.	4J4-F
Difference signal	Variable phase: 1.0 ±0.1 kHz sine wave having a lagging phase angle from the reference phase as specified in Table XII, amplitude in peak-to-peak volts as specified in Table XII and as measured between TP49 (signal) and TP48 (signal return) on the computer.	4J4-G
Audio signal	Sine wave having same characteristics as the sum signal except the amplitude shall be adjustable to 1.0 ±0.1 V peak-to-peak under conditions stated in Step 6 of Table XII.	4J1-E (signal) 4J1-F (signal return)

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3.5.27 Subclutter visibility (SCV) and range rate accuracy test. The computer after having met the requirements of 3.5.1 through 3.5.26 shall meet the subclutter visibility (SCV) and range rate accuracy tests as specified in MIL-R-50651 when substituted into a Radar Set AN/VPS-2. Radar Set AN/VPS-2 and facilities shall be as specified in MIL-R-50651.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Workmanship. The workmanship requirements of MIL-F-13926 shall apply.

3.8 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

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4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three computers for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table XIV and Table XV herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the technical data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

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4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table XIV shall be performed on a 100% basis.

TABLE XIV. Functional tests
(100% Inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major			
101.	Electrical	3.4	4.7.2
102.	Performance	3.5	4.7.3
Minor:		None	N/A

4.5 Special sampling. One computer out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table XIV and shall then meet the tests in Table XV. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.

TABLE XV. Special sampling

CHARACTERISTIC		REQUIREMENTS	TEST PROCEDURE
Critical:		None	N/A
Major			
103.	Vibration	3.3.1	4.7.1.1
104.	Shock	3.3.2	4.7.1.2
105.	Altitude	3.3.3	4.7.1.3
106.	Temperature	3.3.4	4.7.1.4
107.	Humidity	3.3.5	4.7.1.5
Minor:		None	N/A

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of the specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, test equipment listed in Table XVI shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE XVI. Test equipment list

ITEM NUMBER	DESCRIPTION
1. Power supply	6.2 ±0.5 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity, 3 A.
2. Power supply	Dual, 23.0 ±0.25 Vdc supplies, regulated; ac ripple less than 50 mV peak-to-peak; capacity, 1 A.
3. Power supply	Dual, 33.0 ±0.25 Vdc supplies, regulated; ac ripple less than 50 mV peak-to-peak; capacity, 1 A.
4. Power supply	24.0 ±0.5 Vdc, regulated; ac ripple less than 100 mV peak-to-peak; capacity, 100 mA.
5. 60 MHz oscillator	60.0 ±5.0 MHz sine wave, 50 mV output.
6. Variable phase	Provide two output sine wave signals, one as reference phase and one with 0 to 360 Degree adjustable phase lag with respect to the reference phase. Frequency 1.0 ±0.1 kHz. Power output 40 mW (5 V across 600 ohms).
7. Dc digital voltmeter	Visual display 5 digits. Full scale range: 50,000 V or greater. Input resistance: 10 megohms or greater. Accuracy: ±0.01%.
8. Oscilloscope	Dual channel; rise time of scope plus measuring probe, 10 ns/cm or greater; input resistance 8 megohms or greater.
9. Test fixture	See Figure 2. See Table XVIII for test fixture parts list.

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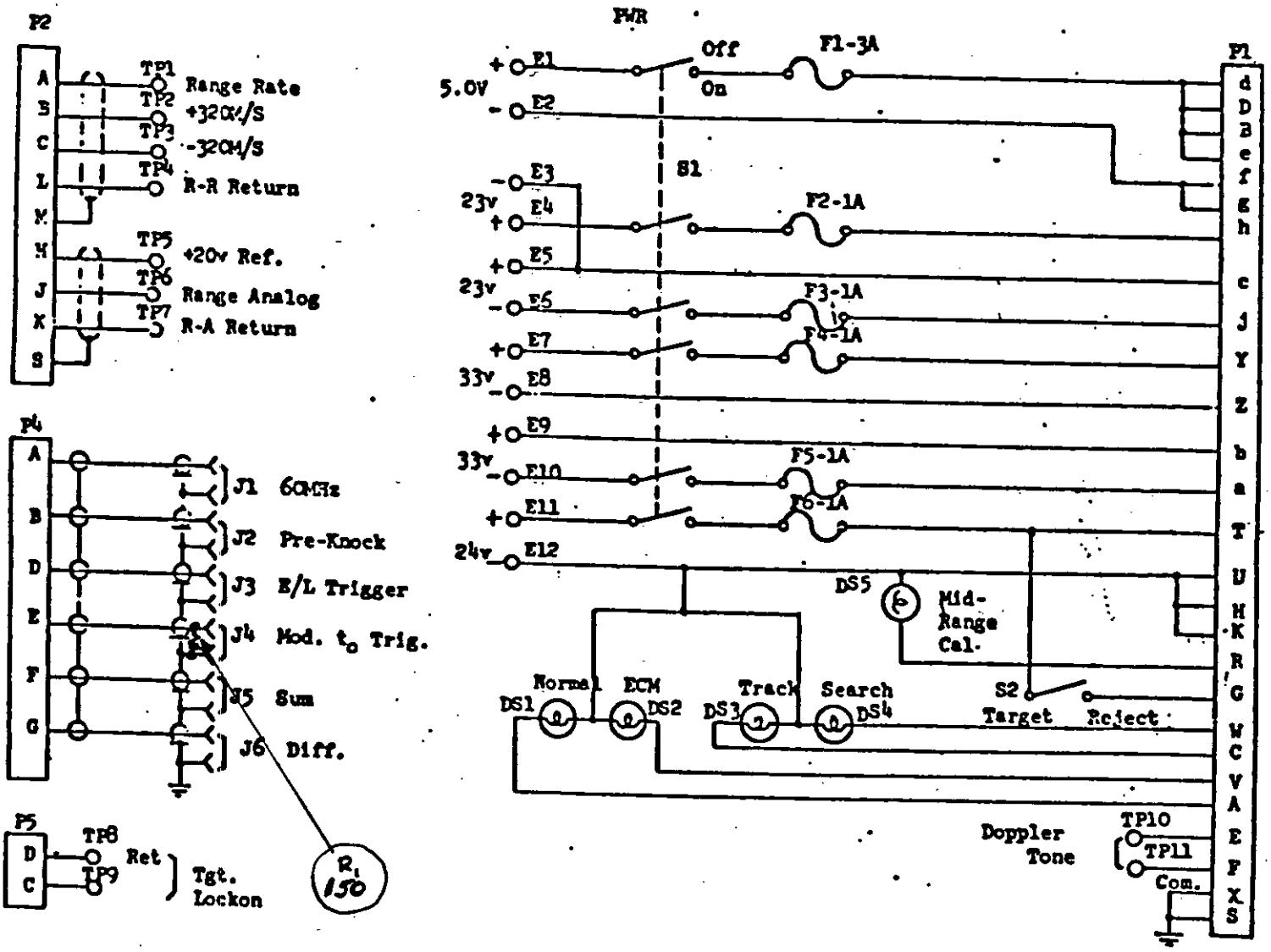


FIGURE 2. Test fixture setup.

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TABLE XVII. Test fixture parts list

DESIGNATION	NOMENCLATURE AND DESCRIPTION
DS1 thru DS4	Lamp, miniature bayonet, type 47.
DS5	Lamp, miniature bayonet, type 313.
E1 thru E12	Binding post, H.H. Smith 257 or equivalent.
F1	Fuse, type 3AG, 3 A.
F2 thru F6	Fuse, type 3AG, 1 A.
J1 thru J6	Connector, Receptacle, UG-625B/U or equivalent.
P1	Connector, Plug, Burndy L18 TF32S6N-A.
P2	Connector, Plug, Burndy L20 TF16P6N-A.
P4	Connector, Plug, Bendix PTSF06SE22-78C.
P5	Connector, Plug, Burndy L12 TF106PX-A.
S1	Switch, 6PST.
S2	Switch, SPST.
TP1 thru TP11	Tip jack, H.H. Smith - 1501 or equivalent
XDS1 thru XDS5	Lamp holder, Dialoc 502-8136-102.
XF1 thru XF6	Fuse holder, Littlefuse 342014.
R1	Resistor, 150 ohm, RL07S151G or equivalent

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, General test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

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4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The computer shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the computer shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the computer shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

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4.7.3 Performance. The selected computers shall be tested as follows: Connect power supplies to the test fixture (Table XVI, item 9) in accordance with Figure 3. Adjust power supplies (Table XVI, items 1 through 4) to deliver input voltages as specified in Table I. Leave the 60 MHz oscillator (Table XVI, item 5) and the variable phase generator (Table XVI, item 6) disconnected from the test fixture (Table XVI, item 9) at this time. For 4.7.3.1 through 4.7.3.26 the requirements of 3.5 shall apply. The Oscilloscope (Table XVI, item 8) shall be used to observe all signals unless otherwise noted. When it is specified to observe signals at test points, it shall mean the test points on the computer unless otherwise noted.

4.7.3.1 Voltage inputs and distribution. Turn the test fixture PWR switch S1 to the ON position and using the digital voltmeter (Table XVI, item 7) observe the distributed voltages at terminals as specified in Table II.

4.7.3.2 Midrange lamp 4DS1. With the conditions of 4.7.3 observe that the midrange lamp 4DS1 shall be dimly lit.

4.7.3.3 Midrange switch 4S1. Using midrange calibration lamp DS5 on the test fixture, depress the midrange switch 4S1 on the computer and check for compliance with 3.5.3.

4.7.3.4 7.5 MHz clock output. Connect the 60 MHz oscillator, item 5, to J1 on the test fixture with the amplitude requirements of 3.5.4. At TP38 observe the 7.5 MHz clock signal as specified in 3.5.4.

4.7.3.5 PRF RESET signal. Sync scope from TP20 and at TP20 observe the PRF RESET signal as specified in 3.5.5.

4.7.3.6 PRF signal. Sync cope from TP20 and observe at TP19 the PRF signal as specified in 3.5.6.

4.7.3.7 PRF cal signal. Sync scope from TP20 and observe at TP19 the PRF calibration signal as specified in 3.5.7 with the midrange calibration switch 4S1 depressed.

4.7.3.8 PRF signal. Sync scope from TP20 and observe at TP31 the PRF signal as specified in 3.5.8 with the midrange calibration switch 4S1 depressed.

4.7.3.9 T1 signal. Sync scope from TP20 and observe at TP21 the T1 signal as specified in 3.5.9.

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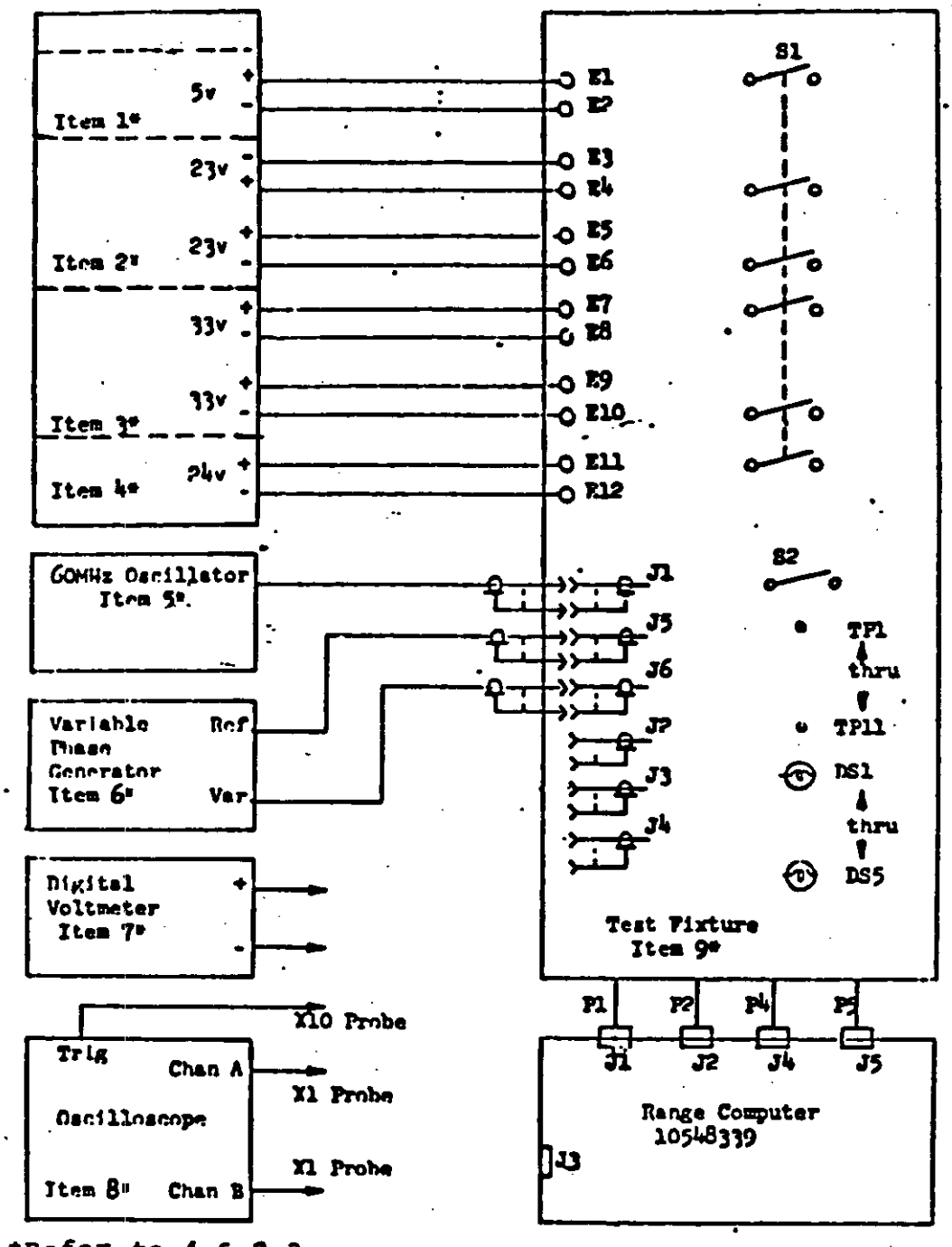


FIGURE 3. Test setup.

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4.7.3.10 MOD To trigger signal. Sync scope from TP31 and observe at TP45 and also at J4 on the test fixture MOD to trigger as specified in 3.5.10.

4.7.3.11 T2 signal. Sync scope from TP21 and observe at TP34 the T2 signal as specified in 3.5.11.

4.7.3.12 T3 signal. Sync scope from TP21 and observe at TP23 the T3 signal as specified in 3.5.12.

4.7.3.13 T11 signal. Sync scope from TP20 and observe the T11 signal as specified in 3.5.13.

4.4.4.14 T1/250 signal. Sync scope from TP9 and observe at TP9 the T1/250 signal as specified in 3.5.14.

4.7.3.15 T3/4KC signal. Sync scope from TP10 and observe at TP10 the T3/4KC signal as specified in 3.5.15.

4.7.3.16 200M signal. Establish the conditions of 3.5.15, using switch S2 on the test fixture to apply the +24 V to 4J1-G and observe the 200M signal at TP11 as specified in 3.5.16.

4.7.3.17 T3 → T10 SERIAL and R SERIAL signals. Sequentially place switches 4S2 through 4S6 as specified in Table III and at TP17 and TP18 observe the T3 T10 SERIAL and R SERIAL signals respectively as specified in Table III and the conditions of 3.5.17.

4.7.3.18 TRACK signals.

4.7.3.18.1 No target condition. With switches 4S2 through 4S6 placed in the positions as specified in 3.5 observe the TRACK signal at TP15 and TP24 as specified in 3.5.18.1.

4.7.3.18.2 Target condition. With the conditions of 3.5.18.2 observe the TRACK signal at TP15 and TP24 as specified in 3.5.18.2.

4.7.3.18.3 Target fade (short term). With the conditions of 3.5.18.3 observe the TRACK signal at TP15 and TP24 as specified in 3.5.18.3.

4.7.3.18.4 Target fade (long term). With the conditions of 3.5.18.4 observe the TRACK signal at TP15 and TP24 as specified in 3.5.18.4.

4.7.3.18.5 Target lock-on signal. With the conditions of 3.5.18.5 observe the target lock-on signal, at TP9 (signal) and TP8 (signal return) on the test fixture, as specified in 3.5.18.5.

4.7.3.19 MID RANGE lamp 4DS1. Observe the conditions of the midrange lamp as specified in 3.5.19.

4.7.3.20 E/L COURSE signal. Observe at TP33 the E/L COURSE signal as specified in 3.5.20.

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4.7.3.21 PRE RANGE trigger signal. Observe at TP36 and also J2 on the test fixture the PRE RANGE (pre-knock) trigger signal as specified in 3.5.21.

4.7.3.22 E/L TRIGGER.

4.7.3.22.1 E/L TRIGGER characteristics. Observe at TP43 and TP44 and also J3 on the test fixture the E/L trigger characteristics as specified in Table IV.

4.7.3.22.2 E/L TRIGGER movements. Sequentially position switches 4S2 through 4S6 as specified in Table V and observe the E/L trigger characteristics at TP43 and TP44 as specified in Table IV and VI.

4.7.3.22.3 E/L TRIGGER leading edges. Position the switches 4S2 through 4S6 as specified in 3.5 and observe the E/L trigger leading edges at TP43, TP44 and J3 on the test fixture, as specified in Table VI.

4.7.3.23 R> +10 M/S signal. Position switches 4S2 through 4S6 as specified in 3.5.23. Sequentially perform the steps of Table VII and observe the R ±10 M/S signal at TP42 as specified in Table VII.

4.7.3.24 Range rate analog signals. Sequentially position switches 4S2 through 4S6 as specified in Table IX and observe the range rate analog ignals, as specified in Table VIII.

4.7.3.25 Range analog signals. Sequentially position switches 4S2 through 4S6 as specified in Table XI and observe the range analog signals at terminals as specified in Table X.

4.7.3.26 Doppler signals response. Connect the variable phase generator, item 6, as shown in Figure 3. Sequentially apply the sum and difference signals as specified in Table XII under the conditions as specified in 3.5.26. Observe that the E/L trigger at TP44 responds as specified in Table XII.

4.7.3.27 Subclutter visibility (scv) and range rate accuracy test. After performing the test of 4.7.3 through 4.7.3.26, substitute the computer into a Radar Set AN/VPS-2 and perform the subclutter visibility and range rate accuracy test as specified in MIL-R-50651. The Radar Set AN/VPS-2 and test facilities used shall be as specified in MIL-R-50651.

4.8 Soldering. Soldering processes shall be inspected or certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.8.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheets PS10548339. The level of protection shall be as specified in the procurement document.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The computer, Range CP-888/VPS-2 is part of Radar Set AN/VPS-2. The computer provides the synchronizing and ranging systems for the radar set. The synchronizing system establishes the PRF of the radar, and generates the timing signals used to coordinate the transmitting, receiving, ranging, and test facilities systems. The ranging system generates the target range and range rate in the form of two dc analog voltages for use in the FCS to compute target motion lead angle.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First Article Sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-OT-90-12009 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
Computer, Range CP-888/VPS-2
Radar Set AN/VPS-2
VADS

6.6 Changes from previous issue.. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A664)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-50685A (AR)	25 May 1990

3. DOCUMENT TITLE
COMPUTER, RANGE CP-888/VPS-2: 10548339

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
US Army ARDEC Standardization & Specification Office	(201) 724-6674	880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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